

1. **Telophase 1:** Choose the incorrect statement:
 - Anaphase lag could create monosomy
 - Anaphase lag results in losing the chromosome
 - Anaphase lag could correct trisomy
 - Anaphase lag can result from improper spindle formation
 - Anaphase lag is genetically determined and regulated
2. **T456-CYTOGENETICS (1):** One of the pathogenic mechanisms that arise from chromosomal abnormalities is position effect, how?
 - Can make a silent gene become active
 - Can convert an active gene into a silent one
 - Can make a gene increase its expression
 - All choices are right
 - Can cause a decrease in gene expression
3. Choose the incorrect statement:
 - Nondisjunction happens in meiosis, not in mitosis
 - Oogenesis happens only in females
 - Nondisjunction can happen in meiosis I and meiosis II
 - Nondisjunction applies to homologous chromosomes separation and sister chromatid separation
 - Sister chromatids can differ after crossing over
4. Choose the incorrect statement about imprinting:
 - Imprinting is stable in cells, but can differ according to tissue type or developmental stage
 - Imprinting needs to be re-programmed with each generation
 - None of the choices
 - Imprinting depends on the parent of origin of the chromosome
 - Imprinting can be exposed by uniparental disomy
5. The most frequent autosomal reciprocal translocation in humans is between chr:
 - 9 and 22

- X and Y
 - 11 and 22
 - 4 and 8
 - 14 and 21
6. One of the pathogenic mechanisms that arise from chromosomal abnormalities is position effect, how?
- Can make a silent gene become active
 - All choices are right
 - Can convert an active gene into a silent one
 - Can make a gene increase its expression
7. Choose the correct statement:
- Most chromosomal abnormalities can be seen at birth stage, rather than that are not as frequent.
 - Most chromosomal abnormalities can be seen in older adults then live birth babies then blastocysts.
 - Aging individuals are expected to have the most of constitutional chromosomal abnormalities in their cells than other life stages.
 - Chromosomal abnormalities, when checked in different individuals of the life of human, have the same frequency.
 - Most chromosomal abnormalities can be seen in blastocysts then live birth babies then older adults.
8. Choose the correct statement:
- Multiple aneuploidies happen in theory but are never seen in humans
 - Sequential nondisjunction happens in both male and female gametogenesis simultaneously
 - Complete nondisjunction will lead to trisomy
 - Simultaneous parental nondisjunction happens in meiosis I and II in the same parent
9. In meiosis, choose the correct statement:
- Chiasma formation, crossing over, disjunction happen in prophase I
 - Disjunction, synapsis, and chiasma formation happen in prophase I
 - Synapsis, chiasma formation, and crossing over happen in prophase I

- All is right
- Crossing over, disjunction, and synapsis happen in prophase I

10. One of the pathogenic mechanisms that arise from chromosomal abnormalities is position effect, how?

- Can make a silent gene become active
- Can convert an active gene into a silent one
- Can cause a decrease in gene expression
- Can make a gene increase its expression
- All choices are right

11. A child patient was found to have an autosomal recessive disease. Mutation testing indicated homozygosity for the pathogenic mutation. Mutation testing in the father found him heterozygous while the mother is wild type for both alleles. How could this happen?

- The child inherited a chromosome from his paternal grandfather and the other homolog from his paternal grandmother
- The child might be trisomic, then trisomic rescue happened and resulted in paternal isodisomy
- The child might be trisomic, then trisomic rescue happened and resulted in maternal heterodisomy
- The child might be trisomic, then trisomic rescue happened and resulted in maternal isodisomy
- The child might be trisomic, then trisomic rescue happened and resulted in paternal heterodisomy

12. Choose the correct statement:

- Aging individuals are expected to have the most of constitutional chromosomal abnormalities in their cells than other life stages
- Most chromosomal abnormalities can be seen at birth stage, earlier than that are not as frequent
- Most chromosomal abnormalities can be seen in older adults then live birth babies then blastocysts
- Chromosomal abnormalities, when checked in different intervals of the life of human, have the same frequency.
- Most chromosomal abnormalities can be seen in blastocysts then live birth babies then older adults

